

JOHN CARLO M. MANALO

Associate Software Engineer | Full Stack Developer | AI/OCR Integration | Power BI Analytics

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PROFESSIONAL SUMMARY

Associate Software Engineer and Information Technology graduate (July 2026) majoring in Business Analytics at Batangas State University. Experienced in full-stack enterprise system development, AI/LLM integration, OCR pipeline development, RAG systems, process automation, and Power BI business intelligence. Independently designed and deployed eight production-level systems at Madison 88 Business Solutions Asia Inc., including an AI-powered AP Invoice Processing Automation system with multi-engine OCR, a SMART Tech Pack-to-Costing Approval Tool with self-hosted LLM and RAG pipeline, and a Power BI Brand Analytics Dashboard delivered in one week for C-level executives. Proficient in React, TypeScript, Node.js, PostgreSQL, Ollama/Qwen 34B, pgvector, Power BI, SQL, and Power Automate.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL, PHP, C++, Java, HTML, CSS

Frontend: React 18, Vite, TailwindCSS, TanStack React Query, Framer Motion, Recharts, React Router

Backend: Node.js, Express, Prisma ORM, JWT, Multer, Zod, Winston, PDFKit, ExcelJS

Database: PostgreSQL (Supabase), MySQL, pgvector, Supabase Real-time Subscriptions, Vector Embeddings

AI & LLM: Google Gemini Vision AI, Anthropic Claude, Groq, Ollama/Qwen 34B, Azure Form Recognizer, RAG Pipeline, pgvector, Prompt Engineering

Cloud & DevOps: Supabase, Netlify, Render, Linux VPS (Hetzner), Nginx, systemd, Git, Turborepo, pnpm

Integrations: Microsoft Graph API, SharePoint, Power Automate, QuickBooks Online API, NextGen API, SFTP, MSAL

BI & Analytics: Microsoft Power BI, DAX, Power Query, Data Modeling, KPI Dashboard Design

Architecture: Monorepo (Turborepo + pnpm), REST API, Role-Based Access Control, Multi-tier Approval Workflows

EDUCATION

Bachelor of Science in Information Technology — Major in Business Analytics **July 2026**
Batangas State University, Lipa City, Batangas

Science, Technology, Engineering, and Mathematics (STEM) **2020 – 2022**
La Purísima Concepcion Academy, Mataas na Kahoy, Batangas

WORK EXPERIENCE

Associate Software Engineer / IT Developer **February 2026 – Present**
Madison 88 Business Solutions Asia Inc. | Alabang, Muntinlupa City, Philippines

- Operated as a one-man development team — solely responsible for the full software development lifecycle of eight enterprise systems, self-directing all project phases: business analysis, requirements gathering from C-level executives, system design, frontend, backend, API integration, database architecture, DevOps, QA, and post-launch support — with no project manager, business analyst, or development team.
- Wore every technical role simultaneously across all projects: Full Stack Developer, Business Analyst, DevOps Engineer, QA Engineer, Data Analyst, Security Engineer, and Solution Architect — independently researching and evaluating technologies, making all architectural decisions, and delivering production-ready systems from zero to deployment.
- Conducted internal Power BI training sessions for company employees — teaching dashboard development, data modeling, DAX, and Power Query to enable data-driven reporting across departments.
- Provided IT and data analytics consulting services within Madison 88 — advising management and department heads on technology solutions, system improvements, and data-driven strategies to support business operations.

AP Invoice Processing Automation System

- Architected an end-to-end AI-powered invoice automation system with a multi-engine OCR pipeline (Google Gemini primary; Groq, Ollama/Qwen 34B, Anthropic Claude, Azure Form Recognizer as fallbacks) and a custom Consensus Extractor with field-level confidence scoring and DSRS graph-based extraction.
- Built Power Automate workflow to monitor SharePoint inbox, capture invoice attachments, and route via SFTP to Linux VPS (Hetzner/Nginx); integrated QuickBooks Online, NextGen API, Microsoft Graph API, and SharePoint for full invoice lifecycle automation with 18 validation rules and multi-tier approval routing.

SMART Tech Pack-to-Costing Approval Tool (In Progress)

- Architecting an enterprise costing workflow automation system featuring Factory CBD digital submission, automated Costing Data Bank updates, guided validation checklists, multi-tier approval routing, and complete audit logging — with self-hosted Ollama/Qwen 34B LLM on Hetzner VPS for AI-powered validation assistance and automated clarification letter generation.
- Building a Historical CBD Intelligence Repository with RAG pipeline using pgvector + Supabase PostgreSQL for semantic Like Styles search, historical consumption/knitting benchmarks, outlier detection, and centralized costing knowledge base. Tech: React, TypeScript, Node.js, Prisma, Supabase + pgvector, Ollama/Qwen 34B, RAG Pipeline, Hetzner VPS, Nginx, Power BI

Brand Analytics Dashboard – Power BI

- Designed and developed an end-to-end Power BI Brand Analytics Dashboard from scratch — directly requested by the company President and Senior Managers — delivered in one week with no existing data pipeline.
- Built data models using Power Query and DAX, and developed a Manual vs. Automation comparison dashboard now used daily by the executive team for strategic decision-making.

Budget Management & Expense Tracker System

- Built a full enterprise Budget Management system with multi-level approval workflow (Employee → Supervisor → Accounting → VP → President) and role-based access control across 9 user roles using React, Node.js, and Supabase PostgreSQL.
- Implemented real-time budget commitment tracking, cash advance management, petty cash reconciliation, PDF/Excel report generation, and a complete audit trail for financial compliance.

IT Headquarters Portal – Centralized IT Hub

- Built a centralized web portal consolidating all internal IT systems under one platform with one-click navigation, real-time project progress tracking, and a built-in Manual vs. Automation comparison dashboard.
- Served as the digital home base for all IT operations, integrating the ITSM system, PO system, Org Chart, and Power BI dashboards under a single interface.

Purchase Order (PO) Cutting System Automation

- Designed and developed an automated PO system integrating Qwen LLM AI for intelligent data extraction and validation — digitizing the entire PO generation, routing, approval, and real-time tracking process company-wide.
- Tech: JavaScript, SQL, MySQL, Qwen LLM, AI Data Extraction, Process Automation, Web Development

Enterprise ITSM Ticketing & Core IT Reporting System

- Built a full enterprise ITSM platform for incident management, service requests, and ticket tracking — utilized by management for SLA monitoring and IT performance reporting company-wide.
- Tech: JavaScript, SQL, IT Service Management, Systems Design, Web Development

Interactive Organization Chart

- Developed a dynamic real-time organizational chart visualizing the full company structure, enabling employees and management to navigate department hierarchies and reporting lines.
- Tech: JavaScript, HTML, CSS, Data Visualization, UI/UX Design

ADDITIONAL EXPERIENCE

Social Media Marketing & Graphic Designer

2024 – 2025

Freelance / Part-time | Remote

- Produced graphic design assets including logos, posters, and promotional materials for various clients while managing multiple concurrent projects remotely.

LEADERSHIP & ORGANIZATIONS

- President — Red Cross Youth Council, BatStateU (2025–2026)
- President — Junior Philippine Computer Society, BatStateU (2024–2025)
- Head of Branding & Creatives — Red Spartan Sports Council, BatStateU (2024–2025)
- Sports Editor — The Apex Publication, BatStateU (2023–2025)

INTERESTS

Full Stack Development | AI/LLM Engineering | Business Intelligence | Process Automation | UI/UX Design | Data Analytics